WOWII: A counterexample to conjecture 138

December 16, 2025

1 Conjecture

If G is a simple connected graph, then $path(G) \geq (2 + u(G)) \cdot dist_{min}(M(G^2))$

Conjecture reference.

Notations

- $path(G)$: The number of vertices of the longest induced path in the graph.
- $u(G)$: 1 if G has a unique maximum independent set, and 0 otherwise.
- $M(G^2)$: The set of vertices of maximum degree in the square of the graph G^2.
- $dist_{min}(M)$: The minimum distance between two distinct vertices of M.

2 Results

2.1 Graph identification

The identified graph is defined by the following graph6 string:

```
XG??Go??CC???C?C??GG?G?H???DG?A?????@??G????GG@??@G
```

2.2 Calculated properties

Property	**Value**
Number of vertices	25
Number of edges	24
Path number $path(G)$	16
Function $u(G)$	1
Vertices $M(G^2)$	$\{6, 23\}$

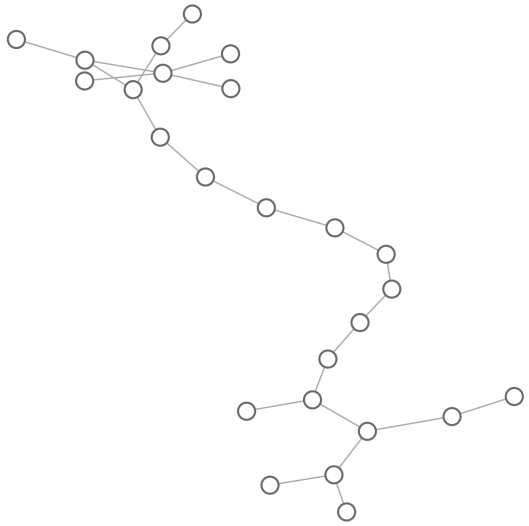


Figure 1: Counterexample

3 Verification of the inequality

There is a potential ambiguity regarding the metric space in which the distance $dist_{min}$ should be calculated. In G according to the literal definition (scenario A), or in G^2 according to the standard convention for power graphs (scenario B). We demonstrate here that the graph refutes the conjecture under **both interpretations**.

3.1 Scenario A: distance in G

According to the provided definition, the distance is measured in the original graph G.

$$dist_{min}(M(G^2)) = dist_G(6, 23) = 11$$

The right-hand side (RHS) becomes:

$$RHS_A = (2 + u(G)) \cdot 11 = (2 + 1) \cdot 11 = \mathbf{33}$$

Verification:

$$16 \geq 33 \quad \Longrightarrow \quad \textbf{FALSE}$$

3.2 Scenario B: distance in G^2

If we consider the distance in the space of the square of the graph G^2:

$$dist_{min}(M(G^2)) = dist_{G^2}(6, 23) = 6$$

The right-hand side (RHS) becomes:

$$RHS_B = (2 + u(G)) \cdot 6 = (2 + 1) \cdot 6 = \mathbf{18}$$

Verification:

$$16 \geq 18 \quad \Longrightarrow \quad \textbf{FALSE}$$

4 Conclusion

Whether one adopts the literal definition or the standard convention, the inequality $path(G) \geq RHS$ is never satisfied for this graph. **This graph is therefore a valid counterexample.**